

Akhilesh Babu Tumati

akhileshbabut24@vt.edu | +1 571-238-0184 | linkedin.com/in/akhilesh-babu-tumati | github.com/AkhileshBabuT

PROFESSIONAL EXPERIENCE

Full Stack Developer (Part-time) | Virginia Tech

June 2025 – December 2025

- Developed a scalable **academic content platform** using **Next.js 15 (React 19)** and **Express.js**, reducing initial FCP from approximately **2.4s to 1.4s (40%)** and supporting sub-second search across 10+ domains.
- Implemented a **serverless cloud infrastructure** on AWS (S3, CloudFront) via **CloudFormation**, selecting serverless to minimize operational overhead and pair cleanly with **MongoDB Atlas**, cutting operational costs by **25%** versus the EC2 baseline.
- Optimized **AWS Lambda** backends and **MongoDB connection pooling**, reducing cold-start latency from **980ms to 540ms (45%)** to handle peak concurrent usage during ingestion windows.
- Hardened auth and access control with **MFA**, **5-tier RBAC**, and **JWT token blacklisting**, securing user sessions across the platform.

Software Engineer | UPS

August 2023 – August 2024

- Orchestrated production deployments for **high-traffic microservices** powering **address analytics** (origin/destination resolution) and **logistics and business analytics** across UPS's global delivery network, shipping releases through **Jenkins CI/CD** pipelines into **OpenShift / Kubernetes** clusters with automated, secure, high-fidelity rollouts.
- Maintained **99.9% system availability** for these distributed services by proactively authoring **8 Grafana alert rules** on **Couchbase cluster** health, reducing P1 SLA incidents during **peak holiday traffic** loads.
- Owned **schema design and data integrity** for production **Couchbase NoSQL** databases backing address and analytics workloads, executing high-fidelity document updates to guarantee consistent reporting across global operations.

Software Developer Intern | Honeywell

January 2023 – July 2023

- Shipped full-stack features for **ASDS (Aerospace Data Services)**, a **customer-facing B2B e-commerce marketplace** for aerospace parts and flight data serving enterprise customers.
- Built reusable **Angular** UI modules backed by **Java Spring Boot** REST controllers, mapping component services through controller endpoints, service layers, and **JPA ORM** to the data layer following **SOLID design principles**, accelerating component reuse.
- Wrote end-to-end **automation test suites** for **payment gateway integrations** and revenue-critical business modules using **Java Selenium** i.e. **Katalon Studio**, reducing manual regression overhead by **30%** and safeguarding transaction flows in production releases.
- Managed **CI/CD pipelines** via **Bamboo** and **Bitbucket** with **Coverity** and **Blackduck** security scans embedded into the build loop, ensuring synchronized enterprise deployments.

TECHNICAL SKILLS

Languages:	Java (Spring Boot), JavaScript (ES6+), TypeScript, Python, SQL, Node.js, Data Structures & Algorithms
Frontend:	React 19, Next.js 15 (App Router), Angular, Tailwind CSS, Tremor UI, Responsive Design, PWA
Backend & APIs:	REST APIs, GraphQL, Express.js, FastAPI, Socket.io, Gemini API, MCP, fal.ai
Cloud & DevOps:	AWS (Lambda, EC2, S3, CloudFront, CloudFormation), Docker, Kubernetes, Jenkins, CI/CD, Bamboo, Git
Databases:	MongoDB (Atlas), PostgreSQL (Prisma ORM), MySQL, DynamoDB, Couchbase, NoSQL
Core Concepts:	Microservices, Distributed Systems, System Design, Schema Design, Scalability, Agile

EDUCATION

Virginia Tech — Master of Engineering in Computer Science

GPA: 4.0/4.0

August 2024 – May 2026

Vellore Institute of Technology — B.Tech. in Computer Science

GPA: 3.5/4.0

August 2023

SELECTED PROJECTS

MCCS Fraud Detection Platform

VT Integrated Product Design (IPD) – MCCS Sponsor, Sep 2024 – Dec 2025

- Engineered a real-time inference pipeline in **Python** using **XGBoost** trained on **GAN-augmented synthetic fraud samples** to address class imbalance, achieving sub-second risk classification across **15,000+ daily transactions**.
- Integrated a **Generative AI assistant** using the **Google Gemini API** to automate **Explainable AI (XAI)** audit reporting, generating **150+ monthly reports** and saving sponsor audit teams **12 hours/week** in manual PDF preparation.
- Developed high-fidelity **React** dashboards using **Tremor** and **Recharts**, surfacing actionable insights into fraud patterns and audit compliance for sponsor stakeholders.

Evol Jewels: AI-Powered Virtual Try-On Kiosk

- Engineered a high-impact retail kiosk using **Next.js 15** and **TypeScript**; integrated **fal.ai** generative workflows to improve virtual try-on image quality by 30% using reasoning-guided generation.
- Awarded 2nd place at Fountane hackathon**; orchestrated dynamic catalog data using **Prisma ORM** and **PostgreSQL** with secure lead capture via QR-code integration and automated email notifications.

PUBLICATIONS

- Tumati, A. B.**, "Face Invariant Classification and Detection of Mythology Characters Using Custom Dataset (ClaDeMuC-CD)." 2023 *IEEE ICAECT*.
- Tumati, A.B.**, "Hybrid ML and DL models for flood level prediction. International Journal of Health Sciences.